Measure sound speed

antocciphysics | TEACHER KEY | CPA

Learning target: Analyze sound travel measurements and their limitations.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 7; 3 course days later.*
+4 mm; partly cancel.

02. Explain a worked example (worked)
Time covers both directions; range=vΔt/2.

03. Look and explain (model)
One-way; 340 m/s.

04. Build the solution (practice)
4 m/s; falls.

05. Use the idea (reason)
Relevant table; device latency/distance calibration/echo path ambiguity accepted.

06. Choose a method (mixed) *Review: Unit 5, Day 10; 12 course days later.*
The larger sphere has 1/4 of the smaller sphere’s area per mass.

07. Exit check (exit)
It uses half the actual sound path; answer half true value.

08. STRETCH: Apply it another way (stretch)
34 and 102 m; separation 68 m.

Teaching note: Investigation companion. Use teacher-approved sound levels. Never label supplied analysis values as measured data.

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01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 7; 3 course days later.*
Resultant -9 mm. Energy also depends on motion, spatial gradients and the surrounding wave field.

02. Explain a worked example (worked)
Time covers both directions; range=vΔt/2.

03. Represent and interpret (model)
One-way; 340 m/s.

04. Solve and justify (practice)
340 m/s; lower 6.8/.0205=331.7, upper 6.8/.0195=348.7 m/s.

05. Analyze the reasoning (reason)
Relevant table; device latency/distance calibration/echo path ambiguity accepted.

06. Choose a method (mixed) *Review: Unit 5, Day 10; 12 course days later.*
Area factor 16, mass factor 64; A/m becomes 1/4. Survival also depends on transfer rates, fragmentation and conditions, not only energy per mass.

07. Exit check (exit)
340 m/s; incorrect 242.9 m/s, biased low.

08. STRETCH: Apply it another way (stretch)
Central 340.9 m/s; bounds 1.5/0.0046≈326.1 to 1.5/0.0042≈357.1 m/s.

Teaching note: Investigation companion. Use teacher-approved sound levels. Never label supplied analysis values as measured data.